

NeoStats Data Engineering Assessment

Retail Sales Data Processing and Business Insights Generation

Submitted By

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Role Applied

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Technology Stack

Python, Pandas, OpenPyXL, Power BI, VS Code

Contents

1	Project Overview	2
2	Problem Statement	2
3	Dataset Description	2
4	Solution Architecture	2
5	Data Model	3
6	Technologies Used	4
7	Data Quality Assessment	4
8	Data Cleaning Strategy	5
9	Data Governance and PII Protection	6
10	Revenue Calculation	6
11	Business KPIs	7
12	Power BI Dashboard	7
13	Business Insights	8
14	Assumptions	8
15	Error Handling	9
16	Scalability Considerations	9
17	Project Deliverables	9
18	Conclusion	9

1 Project Overview

ABC Retail Solutions operates across multiple cities through both online and offline sales channels. The company collects transactional retail data from multiple operational systems. Due to data quality issues such as missing values, inconsistent product names, invalid quantities, and mixed date formats, generating reliable business insights becomes challenging.

The objective of this project is to design and implement an end-to-end Data Engineering pipeline capable of ingesting, cleaning, transforming, validating, and visualizing retail transaction data. The final solution produces a curated analytics-ready dataset and an interactive Power BI dashboard for business reporting.

2 Problem Statement

The goal of this project is to:

- Consolidate retail transaction data from multiple sources.
- Identify and resolve data quality issues.
- Standardize categories, products, and date formats.
- Implement governance controls through PII masking.
- Generate business KPIs.
- Build an interactive Power BI dashboard.

3 Dataset Description

The project uses three worksheets.

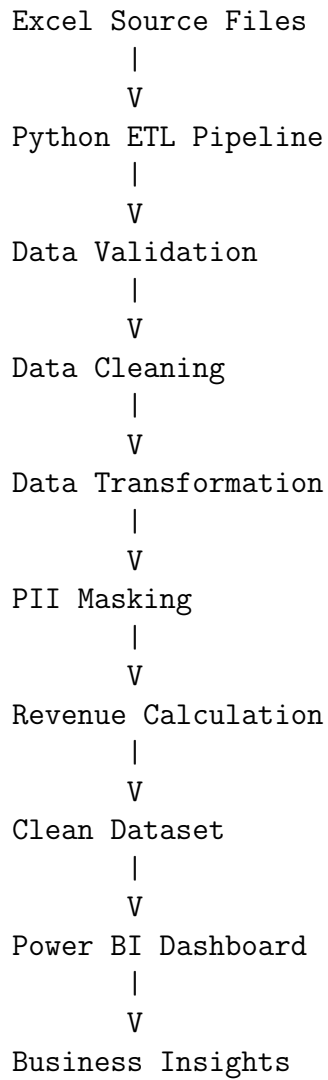
Dataset	Records	Description
product_details	10	Product Master Data
retail_data1	4243	Retail Transactions Source 1
retail_data2	4251	Retail Transactions Source 2

Table 1: Input Datasets

The product master dataset serves as the reference table used to enrich and validate retail transaction records.

4 Solution Architecture

The solution follows a traditional ETL architecture.



The architecture ensures reliable transformation of raw retail data into business-ready information.

5 Data Model

The project follows a simple dimensional model.

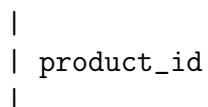
product_details (Dimension Table)

product_id (PK)

product_name

category

price



cleaned_retail_data (Fact Table)

```
-----
transaction_id
customer_id
product_id (FK)
city
transaction_date
quantity
price
discount
revenue
```

The Product Master dataset acts as a dimension table and provides product metadata used to enrich transaction records. The retail transaction dataset acts as the fact table and stores business transactions used for reporting and analytics.

6 Technologies Used

Component	Technology
Programming Language	Python
IDE	VS Code
Data Processing	Pandas
Excel Processing	OpenPyXL
Visualization	Power BI
Storage	Excel
Governance	PII Masking

Table 2: Technology Stack

7 Data Quality Assessment

Several data quality issues were identified.

Issue	Count
Missing Prices	801
Invalid Quantities	93
Duplicate Records	0
Invalid Dates	0

Table 3: Data Quality Issues

Major issues identified:

- Missing price values.
- Invalid quantities (0, -1, -2).
- Category inconsistencies.
- Product naming inconsistencies.
- Mixed date formats.
- Presence of sensitive customer information.

8 Data Cleaning Strategy



```

PS C:\Users\moham\Documents\NeoStats_DataEngineering\Code> python retail_pipeline.py
Data Loaded Successfully

Combined Dataset Shape:
(8494, 16)

Invalid Quantity Records Removed:
93

Missing Prices Before:
801

Missing Prices After:
0

Categories:
['Clothing', 'Electronics', 'Furniture', 'Home Appliances']

Product Names:
['Dining Table', 'Laptop', 'Microwave', 'Mixer Grinder', 'Phone', 'Refrigerator', 'Shirt', 'Shoes', 'Sofa', 'TV']

Date Data Type:
datetime64[us]

Invalid Dates:
0

PII Masking Completed

Revenue Column Created
=====

```

Figure 1: ETL Pipeline Execution Results

The following cleaning activities were performed:

1. Merged retail datasets.
2. Validated duplicate records.
3. Removed invalid quantities.
4. Recovered missing prices using Product ID lookup.
5. Standardized product names.
6. Standardized categories.
7. Converted transaction dates to datetime format.
8. Masked personally identifiable information.

Final dataset size:

- Combined Records: 8494
- Invalid Records Removed: 93
- Final Records: 8401

9 Data Governance and PII Protection

	C	E	F	G	K	M	N	O	Q	R	S	T
1	customer_name	price	product_name	category	quantity	discount	email	phone	year	month	quarter	revenue
2	Troy Mitchell	15000	Mixer Grinder	Home Appliances	3	0.1	Tr***@gmail.com	83*****68	2025	December	Q4	40500
3	Sarah Guerrero	70000	Phone	Electronics	5	0.35	Sa***@yahoo.com	71*****11	2025	July	Q3	227500
4	Samantha Hull	65000	Refrigerator	Home Appliances	2	0.05	Sa***@outlook.com	74*****65	2025	July	Q3	123500
5	Gerald Cooper	80000	Sofa	Furniture	2	0.05	Ge***@gmail.com	97*****01	2025	October	Q4	152000
6	Cameron Black	15000	Mixer Grinder	Home Appliances	1	0.05	Ca***@yahoo.com	99*****77	2025	December	Q4	14250
7	Maria Robinson	3E+05	Laptop	Electronics	5	0.35	Ma***@gmail.com	87*****17	2025	July	Q3	812500
8	Maria Robinson	3E+05	Laptop	Electronics	5	0.35	Ma***@gmail.com	87*****17	2025	July	Q3	812500
9	Charles Brown	3E+05	Laptop	Electronics	2	0.05	Ch***@gmail.com	99*****22	2026	March	Q1	475000
10	Cathy Harris	70000	Phone	Electronics	1	0.05	Ca***@gmail.com	83*****76	2026	February	Q1	66500
11	Linda Lara	45000	Microwave	Home Appliances	3	0.3	Li***@outlook.com	74*****01	2025	August	Q3	94500
12	Jose Thomas	70000	Phone	Electronics	2	0.4	Jo***@outlook.com	83*****63	2025	December	Q4	84000
13	Lisa Cochran	8000	Shoes	Clothing	4	0.3	Li***@gmail.com	83*****19	2025	December	Q4	22400
14	Yvette Clark	80000	Sofa	Furniture	3	0.25	Yv***@gmail.com	60*****34	2025	June	Q2	180000
15	Susan Torres	65000	Refrigerator	Home Appliances	5	0.35	Su***@outlook.com	87*****26	2025	June	Q2	211250
16	Brent Clark	65000	Refrigerator	Home Appliances	5	0.2	Br***@gmail.com	90*****98	2026	February	Q1	260000
17	George Wilkinson	65000	Refrigerator	Home Appliances	5	0.25	Ge***@gmail.com	65*****22	2026	January	Q1	243750
18	Eric Gaines	8000	Shoes	Clothing	5	0.15	Er***@yahoo.com	76*****27	2025	August	Q3	34000
19	Donald Carter	70000	Phone	Electronics	3	0.15	Do***@gmail.com	79*****19	2025	May	Q2	178500
20	Allen Gilbert	15000	Mixer Grinder	Home Appliances	2	0.15	Al***@outlook.com	75*****31	2025	May	Q2	25500
21	Jeremiah Simmons	1300	Shirt	Clothing	4	0.2	Je***@outlook.com	71*****16	2025	September	Q3	4160
22	Angela Weiss	8000	Shoes	Clothing	1	0.15	An***@yahoo.com	64*****39	2025	July	Q3	6800
23	Nicole Williams	65000	Refrigerator	Home Appliances	3	0.05	Ni***@gmail.com	92*****32	2025	December	Q4	185250
24	Kimberly Lloyd	65000	Refrigerator	Home Appliances	4	0.4	Ki***@gmail.com	60*****93	2025	March	Q1	156000
25	Amanda Castillo	15000	Mixer Grinder	Home Appliances	2	0.25	Am***@yahoo.com	92*****48	2026	April	Q2	22500
26	Stephanie Dunn	15000	Mixer Grinder	Home Appliances	3	0.25	St***@gmail.com	74*****44	2025	May	Q2	33750

Figure 2: Cleaned Dataset Showing Standardized Categories, Masked PII, and Revenue Calculation

Sensitive customer information was protected using masking techniques.

Email Example:

Original : johnsmith@gmail.com
Masked : jo***@gmail.com

Phone Example:

Original : 9876543210
Masked : 98*****10

This approach improves privacy while retaining analytical value.

10 Revenue Calculation

Revenue was calculated using the formula:

$$Revenue = Quantity \times Price \times (1 - Discount)$$

This metric was used throughout the reporting layer.

11 Business KPIs

KPI	Value
Total Revenue	Rs. 1,239,132,235
Total Transactions	8401
Unique Customers	1960

Table 4: Business KPIs

Revenue by Category:

Category	Revenue (Rs.)
Electronics	721,682,500
Furniture	257,064,000
Home Appliances	242,328,000
Clothing	18,057,735

Table 5: Revenue by Category

12 Power BI Dashboard

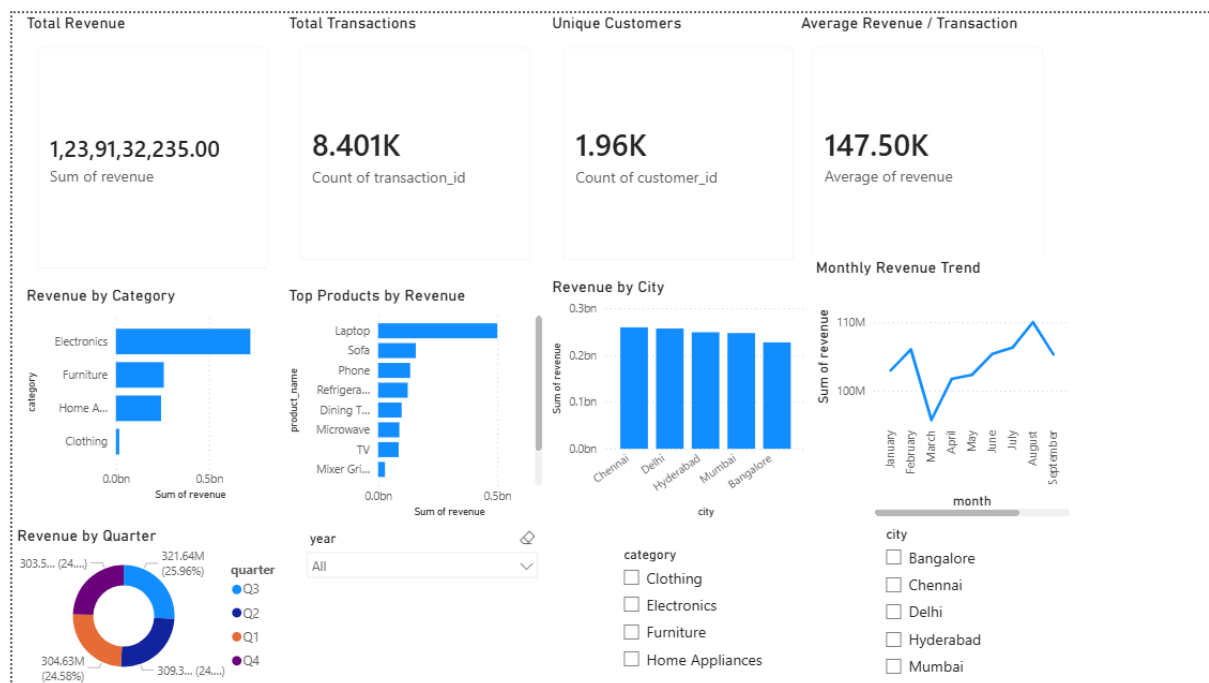


Figure 3: Retail Sales Performance Dashboard

The Power BI dashboard includes:

- Total Revenue KPI

- Total Transactions KPI
- Unique Customers KPI
- Average Revenue per Transaction
- Revenue by Category
- Revenue by City
- Revenue by Quarter
- Monthly Revenue Trend
- Product Revenue Analysis
- Interactive Filters

13 Business Insights

Key findings:

- Electronics generated the highest revenue contribution, accounting for approximately 58% of total revenue.
- Laptop emerged as the highest revenue-generating product across all categories.
- Chennai generated the highest city-wise revenue, followed closely by Delhi and Hyderabad.
- Clothing contributed the lowest overall revenue despite having a comparable transaction volume.
- Revenue trends remained relatively stable across the reporting period, indicating consistent sales performance.
- Furniture and Home Appliances collectively contributed more than 40% of total revenue, demonstrating strong demand in durable goods categories.

14 Assumptions

- Product ID is unique.
- Product Master is the source of truth.
- Quantity less than or equal to zero is invalid.
- Revenue is calculated using discount-adjusted pricing.

15 Error Handling

The ETL pipeline incorporates multiple validation checks to ensure data quality and reliability.

- Detection and removal of invalid quantity values.
- Validation of missing prices and recovery through Product ID lookup.
- Verification of duplicate records.
- Date format standardization and invalid date detection.
- Category and product standardization checks.
- Validation of revenue calculation logic.

16 Scalability Considerations

Although the current implementation uses Excel and Pandas, the architecture can be extended using cloud-native technologies such as Azure Data Factory, Azure Data Lake Storage, Azure Databricks, and Apache Spark. The modular ETL design enables scaling to significantly larger retail datasets.

17 Project Deliverables

The final submission contains:

- Python ETL Pipeline (retail_pipeline.py)
- Cleaned Dataset (cleaned_retail_data.xlsx)
- Power BI Dashboard (NeoStats_Dashboard.pbix)
- Project Documentation Report

18 Conclusion

The project successfully demonstrates an end-to-end Data Engineering solution involving ingestion, validation, transformation, governance, KPI generation, and business intelligence reporting. Data quality issues were resolved through a structured ETL pipeline, resulting in a clean and analytics-ready dataset.

The Power BI dashboard provides actionable business insights and supports data-driven decision making. The solution is scalable, maintainable, and aligned with modern Data Engineering best practices.